

4. [Maximum mark: 7]

A garden centre sells seeds for two varieties of sunflowers: large and giant.

The heights of large sunflowers are normally distributed with mean  $\mu$  and standard deviation  $\sigma$ .

It is known that 25% of large sunflowers have a height of over 180 cm.

(a) Given that  $\mu + A\sigma = 180$ , where  $A \in \mathbb{R}$ , find the value of  $A$ . [3]

The heights of giant sunflowers are also normally distributed.

The giant sunflowers have a mean height that is 35 cm more than that of the large sunflowers and a standard deviation that is double that of the large sunflowers.

It is known that 98% of giant sunflowers have a height greater than 180 cm.

(b) Determine  $\mu$  and  $\sigma$ . [4]

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